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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Data Science for Engineers (course)

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Course outline

Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-08-20, 23:59 IST.

Assignment submitted on 2025-08-20, 18:46 IST

1) Let $f(x) = x^3 + 3x^2 - 24x + 7$. Select the correct options from the following: ***3 points***

- [ ] $x = 2$ will give the maximum for $f(x)$.
- [x] $x = 2$ will give the minimum for $f(x)$.
- [x] Maximum value of $f(x)$ is 87.
- [ ] The stationary points for $f(x)$ are 2 and 4.

Yes, the answer is correct.
Score: 3

Accepted Answers:
$x = 2$ will give the minimum for $f(x)$.
Maximum value of $f(x)$ is 87.

2) Find the gradient of $f(x) = x^2y$ at $(x, y) = (1, 3)$. ***2 points***

( ) $\nabla f = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$

(•) $\nabla f = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$

( ) $\nabla f = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$

( )

(unit?unit=55&lesson=57)
Unconstrained Multivariate Optimization (Continued) (unit?unit=55&lesson=58)
Gradient (Steepest) Descent (OR) Learning Rule (unit?unit=55&lesson=59)
FAQ (unit?unit=55&lesson=60)
Week 4 Feedback Form : Data Science for Engineers!! (unit?unit=55&lesson=156)
Quiz: Week 4 : Assignment 4 (assessment?name=239)
Week 5 ()
Week 6 ()
Week 7 ()
Text Transcripts ()
Download Videos ()
Books ()
Problem Solving

$$\nabla f = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Yes, the answer is correct.
Score: 2

Accepted Answers:

$$\nabla f = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

3) Find the Hessian matrix for $f(x, y) = x^2 y$ at $(x, y) = (1, 3)$ ***2 points***

- $\nabla^2 f = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}$
- $\nabla^2 f = \begin{bmatrix} 3 & 3 \\ 3 & 0 \end{bmatrix}$
- $\nabla^2 f = \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}$
- $\nabla^2 f = \begin{bmatrix} 6 & 3 \\ 3 & 0 \end{bmatrix}$

Yes, the answer is correct.
Score: 2

Accepted Answers:

$$\nabla^2 f = \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}$$

4) Let $f(x, y) = -3x^2 - 6xy - 6y^2$. The point (0, 0) is a ***1 point***

- saddle point
- maxima
- minima

No, the answer is incorrect.
Score: 0

Accepted Answers:
maxima

5) For which numbers b is the matrix $A = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix}$ positive definite? ***1 point***

- $-3 < b < 3$
- $b = 3$
- $b = -3$
- $-3 \leq b \leq 3$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$-3 < b < 3$

Session - July 2025 ()

6) Consider $f(x) = x^3 - 12x - 5$ Which among the following statements are true? ***1 point***

- [ ] $f(x)$ is increasing in the interval $(-2, 2)$.
- [x] $f(x)$ is increasing in the interval $(2, \infty)$
- [ ] $f(x)$ is decreasing in the interval $(-\infty, -2)$
- [x] $f(x)$ is decreasing in the interval $(-2, 2)$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$f(x)$ *is increasing in the interval* $(2, \infty)$
$f(x)$ *is decreasing in the interval* $(-2, 2)$

7) Consider the following optimization problem: ***1 point***

$$\max_{x \epsilon \mathbb{R}} f(x)$$

, where

$$f(x) = x^4 + 7x^3 + 5x^2 - 17x + 3$$

Let x^* be the maximizer of $f(x)$. What is the second order sufficient condition for x^* to be the maximizer of the function $f(x)$?

- ( ) $4x^2 + 21x^2 + 10x - 17 = 0$
- ( ) $12x^2 + 42x + 10 = 0$
- ( ) $12x^2 + 42x + 10 > 0$
- (•) $12x^2 + 42x + 10 < 0$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$12x^2 + 42x + 10 < 0$

8) In optimization problem, the function that we want to optimize is called ***1 point***

- ( ) Decision function
- ( ) Constraints function
- ( ) Optimal function
- (•) Objective function

Yes, the answer is correct.
Score: 1

Accepted Answers:
Objective function

9) The optimization problem $\min_x f(x)$ can also be written as $\max_x f(x)$ ***1 point***

- ( ) True
- (•) False

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

10) Gradient descent algorithm converges to the local minimum **1 po**

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
True